

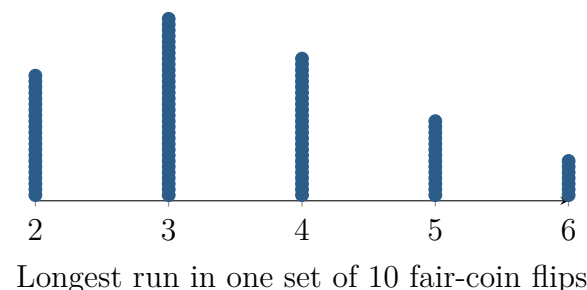
Can Chance Make a Streak?

Unit 2 · Topic 2.3 · TODAY'S PROBLEM

Suppose a coin advertised as fair is flipped 10 times and gives HH-HHHHTTTT. To investigate what chance alone could produce, a class simulates 100 sets of 10 independent fair-coin flips. For each set, the class records the longest run of matching outcomes; the dotplot shows the results.

Claim A: “The coin is rigged—six heads in a row cannot be chance.”

Claim B: “Streaks like this can happen by chance; simulation can tell us how unusual this one is.”



FIRST TAKE (5 min, on your sheet)

Which claim seems better supported right now? Cite one detail from the sequence or dotplot.

- DOK 1** (a) State the longest run in the observed sequence and the most common longest-run value among the simulated sets.
- DOK 2** (b) Use the dotplot to estimate the probability that 10 fair-coin flips have a longest run at least as long as the observed run. Interpret the estimate in context.
- DOK 3** (c) Which claim is better supported by this evidence? Cite the simulation result that settles your choice. Then describe a larger simulation that would more precisely settle the disagreement: state what one trial is, what to record and count, how many trials to use, and what result would make you doubt that the coin is fair.

Video + follow-along: u4_lesson1-2_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

